MECHANICAL & AEROSPACE ENGINEERING



Tejaruthwik Chataboina

CAD & FEA Engineer · Additive Manufacturing · M.Eng. Mechatronic & Cyber-Physical Systems

2.5 years designing and validating precision aerospace hardware for GE aircraft engine programmes, now in the thesis phase of a Master's in Mechatronic and Cyber-Physical Systems. Working at the intersection of 3D CAD, structural/thermomechanical FEA, topology optimisation and additive manufacturing.

📍 Cham, Germany 🎓 M.Eng. Mechatronic & Cyber-Physical Systems, THD

🛠 Siemens NX · ANSYS APDL · HyperMesh

ABOUT

Background

I'm a mechanical and aerospace engineer currently completing a Master's in Mechatronic and Cyber-Physical Systems at Technische Hochschule Deggendorf, Campus Cham — now in the thesis phase. Before moving to Germany, I spent 2.5 years at Quest Global Engineering Services as a client-side design and integration engineer for IHI Corporation, supporting GE aircraft engine programmes (GEnx-1B, GEnx-2B, Passport-20, GE9x) with 3D CAD, structural and thermomechanical FEA, and hands-on assembly, integration and test of precision aerospace hardware.

Since starting my Master's, I've worked on two team case studies that carry that same CAD-and-FEA mindset into new domains: a topology-optimised, additively manufactured robotic gripper, and the mechanical design and simulation of a vision-guided autonomous ground vehicle. Both are detailed below, together with my specific role on each.

LOCATION
Cham, Bavaria, Germany

CURRENT FOCUS
M.Eng. thesis phase, THD Campus Cham

PROFESSIONAL BACKGROUND
2.5 years, GE engine programmes via Quest Global / IHI

CORE TOOLS
Siemens NX, ANSYS APDL, HyperMesh, SolidWorks

LANGUAGES
English (fluent), German (A2, B1 in preparation), Telugu (native)

PROFESSIONAL EXPERIENCE

Industry work

Mechanical Design & Integration Engineer

Quest Global Engineering Services · client-side at IHI Corporation

Aug 2022 – Feb 2025 · Bangalore, India

- Created and maintained 3D CAD models, part drawings and assembly documentation in Siemens NX for propulsion hardware on GE aircraft engine programmes — GEnx-1B, GEnx-2B, Passport-20 and GE9x.
- Carried out structural and thermomechanical FEA in ANSYS APDL and HyperMesh on engine components — defining load cases, boundary conditions and contact interfaces, and documenting results to support engineering decisions.
- Took part hands-on in Assembly, Integration and Test (AIT) of precision aerospace hardware: prototype assembly, test-article preparation and functional validation.
- Supported non-conformance analysis through the Material Review Board (MRB), assessing aerospace alloys against design specifications and writing engineering dispositions.
- Coordinated with cross-functional teams (thermal, quality, manufacturing) and with IHI's engineering teams in Japan across several concurrent programmes.
- Received the **"On the Fly" Award** twice (March 2024) for design and integration quality.

`Siemens NX` `ANSYS APDL` `HyperMesh` `AIT` `MRB / Non-conformance`

Programme and process details are described at a general level only, in line with customer confidentiality — no drawings, part numbers or customer data are shown.

CASE STUDIES

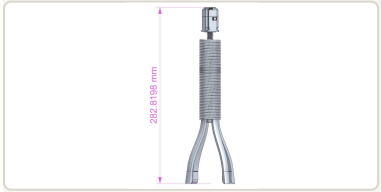
Academic Projects

Two team case studies from my Master's programme at THD, chosen because they best show CAD, FEA and additive-manufacturing skills relevant to design and simulation roles. Team size and each member's role are stated explicitly.

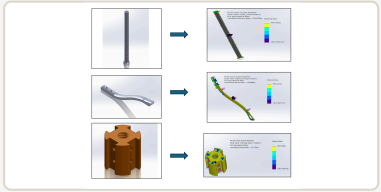
Topology-Optimised Robotic Gripper

Team case study · 5 students

Design, structural validation and additive manufacturing of a lightweight, servo-actuated 4-finger gripper for 5 kg cubic payloads.



Gripper assembly — CAD model showing the servo/lead-screw drive, spring body and four-finger claw.



Topology optimisation of the center screw, finger and claw-holder — solid baseline (left) vs. optimised geometry (right).

PROBLEM / GOAL

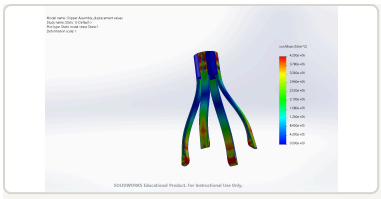
Industries handling small, dense, cube-shaped parts need a lightweight, low-tooling-cost gripper that can be adapted to different cube sizes. The brief was to design, optimise and additively manufacture a claw-type gripper that safely and repeatably handles cubic objects up to 5 kg.

MY ROLE

Design Engineer on a 5-person team: created the 3D CAD design of the gripper and ran the topology optimisation on the claw, center screw and claw-holder; updated design drawings based on functional-test feedback.

TOOLS & METHODS

`SolidWorks CAD` `SolidWorks Simulation (static FEA)`



Static FEA validation — von Mises stress on the optimised claws under the design load case.

`SIMP topology optimisation` `FDM (TPU-95A / PLA)` `Bambu Lab H2S` `Cost / break-even modelling`

APPROACH

Designed a servo/lead-screw-driven 4-finger claw mechanism, where axial motion of a central claw-holder compresses a spring body that closes the flexible claws around the object. Started from a solid, conservatively-sized baseline claw, applied the two governing load cases (normal gripping force and vertical shear from the payload weight), then ran SIMP-based topology optimisation on the claw, center screw and claw-holder to remove low-stress material while keeping the functional interfaces untouched. The optimised geometry was re-checked with static FEA, cleaned up for FDM manufacturability, and printed in TPU-95A (flexible claws/spring) and PLA (rigid housing) — TPU-95A was chosen over softer TPU variants for its higher stiffness, easier printability, and roughly 67% lower material cost than TPU-70A.

KEY RESULTS

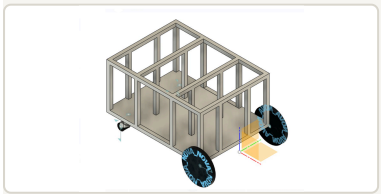
- 33% mass reduction versus the solid baseline design, with stresses kept within safe limits
- Handles cubic objects up to 5 kg with stable, slip-free gripping and elastic recovery after release, confirmed in physical testing of the printed gripper
- Estimated manufacturing cost of ≈€180 per unit on a 4-printer FDM cell, with a break-even point of ≈160 units

What I learned — translating a topology-optimisation output into a part that is actually printable and assemblable meant re-adding manufacturing constraints (minimum wall thickness, support access, functional interfaces) by hand rather than trusting the raw optimiser geometry. Running the CAD and optimisation work inside a 5-person team also meant keeping drawings traceable as the design changed in response to test feedback.

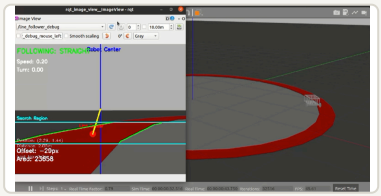
Vision-Guided AGV — Chassis Design & Simulation

`Team case study · 4 students · Grade 1.0`

Mechanical platform and URDF model for a differential-drive AGV, validated in a closed-loop ROS/Gazebo simulation with LQR control and OpenCV-based lane detection.



AGV chassis — CAD model later converted to URDF for Gazebo/Simscape integration.



Closed-loop simulation — OpenCV lane detection (left) driving the AGV along a circular path in Gazebo (right).

PROBLEM / GOAL

Design, model and simulate a control and sensing system for an Automated Guided Vehicle (AGV) that follows a marked path autonomously — covering the mechanical platform, the vehicle dynamics and controller, and camera-based lane detection, integrated in one Gazebo simulation.

MY ROLE

Mechanical CAD design and URDF conversion for the AGV chassis; authored the report's abstract, conclusion and results sections. Teammates derived the state-space dynamics, designed and compared the controllers, and built the OpenCV/ROS lane-detection pipeline.

TOOLS & METHODS

`Fusion 360 (CAD)` `SolidWorks → Simscape Multibody` `URDF` `ROS Noetic` `Gazebo 11` `MATLAB / Simulink` `OpenCV`

APPROACH

Designed a differential-drive chassis — two powered front wheels, a pair of free-turning rear castors, and an aluminium cargo tray on top — modelled with defined joints and linkages between wheels and base link. Converted the CAD assembly into a URDF so the same geometry could drive both a Simscape Multibody dynamic validation and the full Gazebo simulation. That shared chassis model is what let the team's separately-developed LQR controller and OpenCV lane-detection pipeline be tested together as one closed-loop system.

KEY RESULTS

- Case study graded 1.0 (top grade) at THD
- Open-loop system confirmed unstable (eigenvalues [0, 0.8, 0]) and cross-coupled (RGA = [[0.5, 0.5], [0.5, 0.5]]), establishing the need for a MIMO controller
- Of four controllers compared (Pole Placement, LQR, Mixsyn, Loopsyn), LQR gave the lowest overshoot — 0.89 rad on heading — and the closest setpoint tracking
- Simscape validation: at a commanded 0.1 m/s, the AGV reached a 90° heading change in ≈16.6 s
- Full system in Gazebo: HSV/OpenCV lane detection plus the LQR controller aligned the robot to the path centre in ≈15 s

What I learned — a URDF is only as useful as the CAD decisions behind it — joint placement and link naming in the original assembly directly determined how clean the Simscape and Gazebo integration turned out to be. Working across mechanical, controls and computer-vision teammates also meant agreeing on a shared coordinate convention early, instead of reconciling it after each piece was already built.

SKILLS

Technical Toolkit

CAD & DESIGN

- Siemens NX (2.5 yrs, professional)
- SolidWorks
- Fusion 360
- Engineering drawings, BOM, GD&T

FEA & SIMULATION

- ANSYS APDL (professional)
- HyperMesh
- SolidWorks Simulation
- Topology optimisation (SIMP)

ADDITIVE MANUFACTURING

- FDM process & material selection (TPU, PLA)
- Design for Additive Manufacturing (DfAM)
- Manufacturing cost / break-even modelling

ROBOTICS & CONTROLS

MATLAB / Simulink, Simscape Multibody

ROS Noetic, Gazebo, URDF

OpenCV

QUALITY & INTEGRATION

Assembly, Integration & Test (AIT)

MRB / non-conformance analysis

Cross-functional & cross-site coordination

LANGUAGES

English — fluent

German — A2, B1 in preparation

Telugu — native

EDUCATION

Academic Background

M.Eng. Mechatronic and Cyber-Physical Systems

Technische Hochschule Deggendorf (THD), Campus Cham

`Mar 2025 – Present · thesis phase`

Coursework: Additive Manufacturing, CAD Design, Advanced Modelling & Simulation, Cyber-Physical Systems, Advanced Robotics, ROS, Human-Machine Interfaces (VR/AR), Functional Safety.

B.Tech. Mechanical Engineering

TKR Engineering College, affiliated with JNTU Hyderabad

`Jun 2018 – Jun 2022`

Core subjects: FEM, Thermodynamics, Fluid Mechanics, Metallurgy, Theory of Machines, Manufacturing Processes.

RECOGNITION

Awards

🏅 **"On the Fly" Award**

IHI Corporation / GE9x programme — awarded twice, March 2024, for design and integration quality.

TRAINING

Hands-On Additive Manufacturing

Advanced Manufacturing Technologies — Summer School

University of West Bohemia, Faculty of Mechanical Engineering

`Feb 2026 · Pilsen, Czech Republic`

Hands-on operation of industrial LPBF metal and FDM printers — produced 3 physical parts end-to-end from CAD to finished component. Covered topology optimisation, metals-vs-polymers process selection, support strategy and post-processing. Included an industrial visit to a large-format robotic additive manufacturing line (CEAD pellet-based system) at a Czech manufacturing facility.



LPBF metal print produced during the hands-on sessions.



Industrial visit — CEAD large-format robotic AM system.



Cohort outside the Faculty of Mechanical Engineering, Pilsen.

GET IN TOUCH

Contact

EMAIL

tejaruthwikyadavchataboina@gmail.com

PHONE

+49 15510140104

LINKEDIN

linkedin.com/in/tejaruthwik-yadav

ADDRESS

Holzgartenstraße 4, 93413 Cham, Bavaria, Germany

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